ÖMER FARUK KOÇ

AI Platform & MLOps Engineer

Sivas, Türkiye • +90 507 219 2310 • omerfkoc98@gmail.com
linkedin.com/in/omerfkoc • github.com/negativexq • omerfkoc.dev

SUMMARY

AI Platform and MLOps Engineer with **3+ years of production experience** building and operating ML and Generative AI systems in financial services. Experienced in Kubernetes-based model serving, ML lifecycle management, on-premises LLM infrastructure, and production AI workflows. Combines enterprise operations experience with hands-on engineering of **AI platform control planes, RAG and agentic systems**, observability, evaluation, and reliability.

PROFESSIONAL EXPERIENCE

MLOps & Analytics Engineer **Mar 2023 – Mar 2026**
Fibabanka — Analytics Center of Excellence *Istanbul, Türkiye*

- Owned deployment and lifecycle management for **10 production ML models** on Iguazio/MLRun, Docker, and Kubernetes, covering validation, versioning, promotion, retraining, monitoring, and containerized FastAPI model serving; contributed to **5 production LLM workflows**.
- Re-architected sequential Oracle feature-engineering workflows into parallel **dbt + Apache Airflow** pipelines, reducing end-to-end processing from **2–3 hours to 25–30 minutes** while improving dependency management, maintainability, and operational visibility.
- Built a **centralized feature platform** and reusable training, inference, and analytics pipelines across Oracle, Trino, and SQL Server, replacing duplicated feature logic with shared definitions, validation controls, and production data contracts.
- Prototyped AI workloads in Azure and migrated validated services to **on-premises GPU infrastructure**, deploying quantized open-source LLMs and internal inference services under banking data-residency and security constraints.
- Engineered a **call-center intelligence pipeline** combining Whisper-based transcription, speaker diarization, LLM analysis, structured KPI extraction, data-quality validation, and operational monitoring.
- Developed RAG and document-intelligence workflows integrating embeddings, retrieval, LLM processing, structured outputs, and API-based integration into internal AI applications.

TECHNICAL EXPERTISE

Programming & Backend: Python, Go, SQL, FastAPI, Pydantic, REST APIs, PostgreSQL, Redis, React, TypeScript

AI Platform & MLOps: Kubernetes, Docker, MLRun, MLflow, KServe, Knative, Argo Workflows, Helm, Argo CD, vLLM, Ollama, GPU/LLM Serving

Generative AI & Agentic Systems: PydanticAI, LangGraph, LangChain, Agentic RAG, Hybrid Retrieval, Qdrant, Reranking, Multimodal Document Understanding, Hugging Face, PyTorch, QLoRA, LLM Evaluation, Guardrails

Infrastructure & Observability: Linux, GitOps, GitHub Actions, Terraform, OpenTelemetry, Langfuse, Jaeger, Prometheus, Grafana, Loki, Tempo, Airflow, dbt, Kafka, Azure

SELECTED ENGINEERING PROJECTS

ML Platform Infrastructure [GitHub]
Architected a **Kubernetes-native ML platform** with a PostgreSQL-backed control plane, reconciliation loops, and a separate **Go inference gateway**. Integrated Argo Workflows, MLflow, KServe, and Knative for training, model lifecycle, GPU/LLM serving, and automated deployments. Implemented multi-tenant isolation, versioned data lineage, scheduling, model monitoring, and **health-gated canary rollbacks**. Engineered GitOps delivery, observability, security controls, and failure-recovery mechanisms with integration and fault-injection testing.

Agentic SRE [GitHub]
Architected an **evidence-driven Kubernetes incident investigation platform** integrating Prometheus, Loki, Tempo, Kubernetes Events, and workload topology to reconstruct failure timelines and investigate root causes. Engineered a **deterministic RCA engine** with bounded read-only investigation, evidence correlation, immutable evidence records, and replayable decisions. Built a least-privilege, watch-based cluster connector using **gRPC/mTLS**, with auditable access, explicit uncertainty handling, and controlled resource consumption.

Agentic Customer Service Platform [GitHub]
Engineered a **stateful Agentic AI execution platform** using LangGraph, FastAPI, and PostgreSQL, separating LLM-generated proposals from deterministic execution authority. Designed **durable workflows, checkpoint recovery, and authorization boundaries**, with policy enforcement, human confirmation, and idempotent operations. Implemented RAG integration, failure recovery, replay protection, and auditable execution trails. Instrumented agent and backend workflows with **OpenTelemetry**, privacy-aware tracing, and resilience testing across failure scenarios.

Knowledge Base RAG [GitHub]
Architected an **end-to-end agentic document research platform** using **PydanticAI, typed ResearchState, and bounded tool orchestration** for multi-step reasoning across complex PDFs, tables, and charts. Integrated hybrid retrieval, multimodal evidence extraction, source-grounded calculations, and independent semantic validation with tenant-isolated access controls. Achieved **85.8% verified outcomes across 120 MMLongBench-Doc questions** through iterative evaluation. Built an **interactive React research workspace** featuring live investigation timelines, visual source inspection, transparent calculation breakdowns, background research, and shareable reports.

EDUCATION

B.Sc. Electronics Engineering, Gebze Technical University